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## PERFORMANCE COMPARISON OF MULTISTABLE CONFIGURATIONS IN A COMPACT ENERGY HARVESTER

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**Abstract:** Energy harvesting technology using smart structures have been developed since the 1990's. Over the years, a variety of transduction mechanisms have been explored, particularly those based on smart materials such as piezoelectric and magnetostrictive elements. Despite the transducer type, the structure design of the harvester plays a crucial role in its overall performance. The introduction of nonlinearities, such as magnetic interactions, can increase the number of stable equilibria in the system, potentially leading to significant enhancements in output power and operational bandwidth. Moreover, compact designs are essential for enabling applications in confined spaces. This work deals with the performance investigation of a compact dual-beam mechanical energy harvester exhibiting multistable characteristics. The multistability is induced by magnetic interactions, which can be tuned by adjusting the spatial arrangement of the magnets within the system. The harvester's performance is evaluated under different stability conditions. Results indicate that increasing the number of stable equilibria generally broadens the system bandwidth, enhances robustness in power conversion at higher frequencies, and reduce complexity of dynamical behavior. However, it also leads to a substantial decrease in maximum output power, representing a trade-off in energy conversion efficiency. Further investigation is required to determined whether these trends are configuration-specific or reflect a more general behavior of such systems.

**Keywords:** Energy Harvesting, Nonlinear Dynamics, Smart Materials and Structures, Multistability, Compact Structures

### 1. INTRODUCTION

The continuous evolution of semiconductor technology has significantly reduced the power demand of electronic devices (Zhu *et al.*, 2020). This trend enables the use of ambient mechanical energy sources, such as vibration, sound, wind, sea waves, river flows, and biomechanics, to power low-power electronics (duToit *et al.*, 2005), small-scale systems (Toprak and Tigli, 2014), or charge conventional batteries (Daqaq *et al.*, 2020).

To harvest ambient mechanical energy, transducers such as electromagnetic converters, triboelectric structures, and smart materials are widely used (Safaei *et al.*, 2019; Liu *et al.*, 2021). These devices exhibit multiphysical coupling, converting mechanical energy into electrical energy. They can be attached to a host structure that acts as an energy-transfer path between the environment and the transducer, and each technology has distinct advantages and limitations.

In this context, smart materials are often preferred for applications requiring compactness. Piezoelectric materials are the most widely reported in the literature because of their high electromechanical coupling coefficients and ease of fabrication in different geometries (Kim *et al.*, 2015). Magnetostrictive materials can also exhibit high coupling coefficients and do not suffer from depolarization or aging; however, they can be difficult to integrate with microelectronic devices because they require bulky pickup coils (Deng and Dapino, 2017). Magnetic shape memory alloys have also been reported, although relatively few studies have addressed their use in energy harvesting (Karaman *et al.*, 2007).

The harvester structure is a key design factor in energy harvesting devices, as structural configuration directly influences performance. The archetypal configuration is the cantilever beam, which is simple to construct, compact, and exhibits resonator-type behavior (Erturk and Inman, 2009). Its main limitation is reduced efficiency when the excitation frequency deviates from the natural frequency. Under these conditions, beam deflection decreases, resulting in low electrical output and restricting applications to scenarios in which the environmental excitation frequency is known (Zhang *et al.*, 2018). This limitation has motivated researchers to develop improved designs.

The evolution of cantilever-based harvesters has introduced additional mechanical degrees of freedom (DoF), which can improve performance by creating more efficient operating regions (Ou *et al.*, 2012). This development led to the design proposed by Wu *et al.* (2014), which comprises an outer beam and an inner beam. This configuration brings the first two natural frequencies closer together, thereby enlarging the operating region and making the 2-DoF cantilever-based

system more compact.

Additionally, introducing nonlinear characteristics into energy harvesting systems has been shown to improve performance. Nonlinear harvesters can provide broadband response, making them suitable for general applications. In this context, multistable energy harvesters are nonlinear devices with multiple stable positions, a feature that can substantially enhance performance. Multistability can be achieved through several techniques, such as magnetic interactions (Stanton *et al.*, 2010) or mechanically or magnetically induced buckling (Osinaga *et al.*, 2022; Zhu and Zu, 2013).

Researchers have extensively investigated multistability in one-DoF systems, including bistable (Costa *et al.*, 2021), tristable (Kumar *et al.*, 2015), tetrastable (Zhou *et al.*, 2017b), and pentastable (Zhou *et al.*, 2017a) configurations. A key challenge in increasing the number of stable positions is the formation of potential energy barriers, which may reduce performance depending on the excitation level. In general, the literature indicates that increasing the number of stable positions can mitigate this barrier.

Recently, an enhanced dual-beam design was proposed using magnetic interactions and transducer placement on each beam (Costa and Savi, 2024; Costa *et al.*, 2024). In the present work, this design is extended by exploring different magnetic configurations for each DoF. A general potential energy formulation based on a polynomial series is proposed to model beam-magnet interactions. The results reveal unprecedented multistability configurations, with up to 16 stable equilibria (16-stable). The performance of each configuration is then evaluated and compared.

## 2. DESIGN CONCEPT AND THEORETICAL MODEL

This section presents the expanded design concept of the general multistable compact energy harvester, as well as detailing the underlying assumptions considered for its modeling.

A general structure for the compact multistable energy harvester is presented in Figure 1. The design is composed of a cut-out outer beam, leaving space available for an inner beam. Each beam has a tip mass, an attached piezoelectric element, and magnetic interactions formed by a customizable set of magnets around the tip mass. Different types of stability conditions can be achieved depending on the magnetic setup. Specifically, Figures 1(a), 1(b), and 1(c) show examples of different configurations that can manifest up to 4, 9, and 16 stable equilibrium positions, respectively.

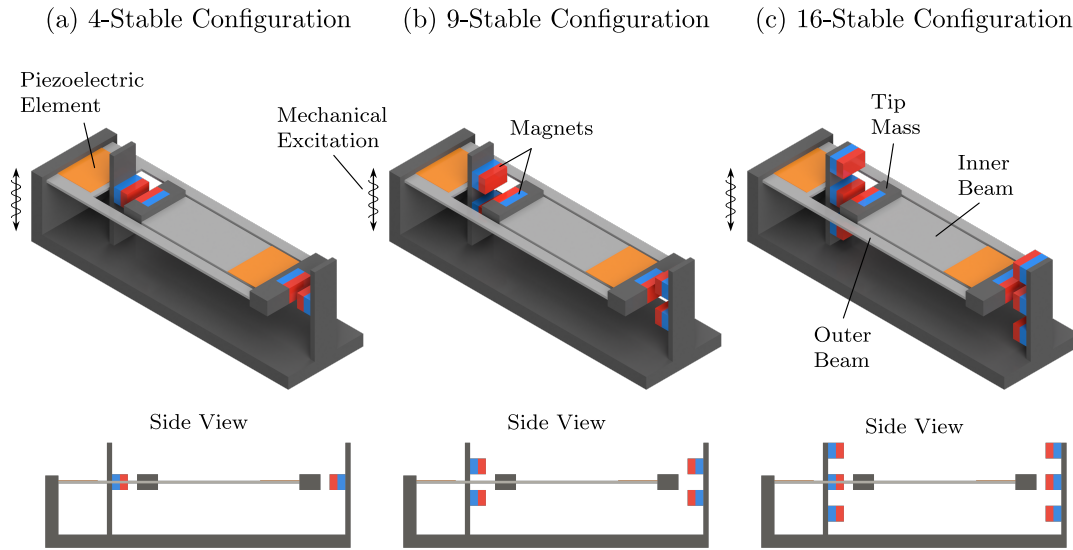


Figure 1: **Design concept of the Compact Multistable Energy Harvester.** Magnet placement can be modified to achieve (a) Up to 4 stable configuration, (b) up to 9-stable configuration, and (c) up to 16-stable configuration.

### 2.1 Physical Modeling

Consider the representation of the Compact Multistable Energy Harvester by a 2-DoF electromechanical lumped model as shown in Figure 2, where each beam is represented by a 1DoF. Considering that the subscript  $i = 1, 2$  denotes the properties associated with each one of the two degrees-of-freedom, structural properties are represented by  $m_i$ ,  $c_i$ ,  $k_i$ , which represents the equivalent mass, equivalent mechanical dissipation, the equivalent linear stiffness of the beam structure. A piezoelectric element is associated with each beam, and can be represented by an electromechanical coupling coefficient,  $\theta_i$ , an internal equivalent capacitance,  $C_{p_i}$ , and an internal resistance  $R_{p_i}$ . An electrical circuit is attached to the piezoelectric elements, and are assumed to be simple resistive elements, each with a load resistance  $R_{l_i}$ . The output voltage of each circuit is represented by  $v_i(t)$ , and the equivalent electrical resistance in each circuit,  $R_i$ , is composed by the piezoelectric internal resistance and the load resistance connected in parallel, such that  $R_i = R_{l_i} R_{p_i} / (R_{l_i} + R_{p_i})$ .

The system is subjected to a support excitation  $z_b(t) = A \sin(\omega t)$ , where  $A$  and  $\omega$  are the excitation amplitude and frequency, respectively. This excitation induces the displacements  $z_i(t)$  in each DoF. Furthermore, the displacement of each mass are associated with the displacements of the real harvester's structure as depicted in Fig. (2c). When  $z_1(t)$  is positive, the motion of the outer beam is directed upward, whereas when  $z_2(t)$  is positive, the motion of the inner beam presents a downward direction (Krishnasamy *et al.*, 2018). Finally, the upper dot represents derivatives with respect to time, as  $\dot{\square} = d\square/dt$ , and the effects of gravity are neglected.

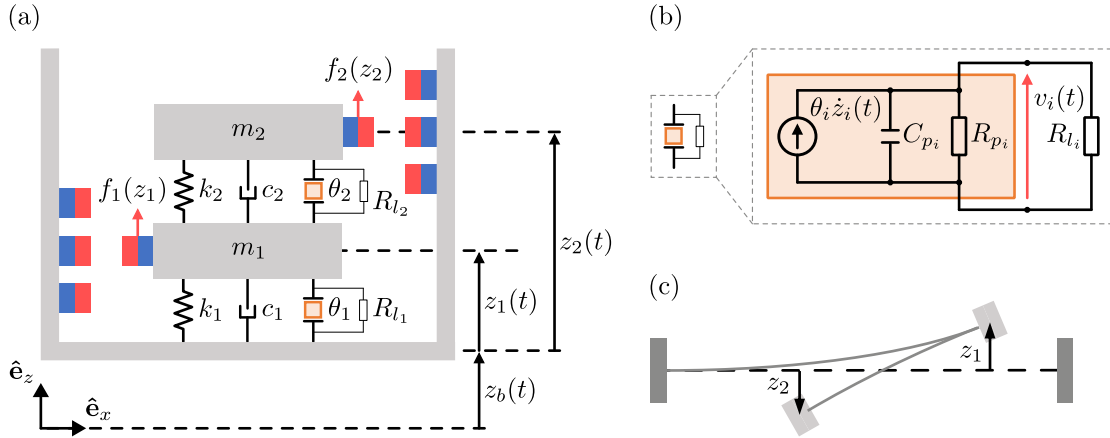


Figura 2: **Lumped electromechanical model representing (a) the general configuration of the compact multistable energy harvester, (b) the equivalent electrical circuit in each DoF, composed by the piezoelectric element and with an attached resistive load element, and (c) the respective positive  $z_i$  directions in the original structure.**

The electromechanical equations are derived using Euler-Lagrange equations with two mechanical generalized coordinates ( $z_1(t)$ ,  $z_2(t)$ ) and two electrical generalized coordinates ( $\psi_1(t)$ ,  $\psi_2(t)$ ). While  $z_1(t)$  and  $z_2(t)$  are displacements,  $\psi_1(t)$  and  $\psi_2(t)$  are magnetic flux linkages, such that  $\dot{\psi}_i(t) = v_i(t)$ , ( $i = 1, 2$ ). Thus, with  $\mathbf{u} = [z_1(t), z_2(t), \psi_1(t), \psi_2(t)]$ , the governing equations are obtained from Equation (1).

$$\frac{d}{dt} \left( \frac{\partial \mathcal{L}}{\partial \dot{u}_j} \right) - \frac{\partial \mathcal{L}}{\partial u_j} + \frac{\partial D}{\partial \dot{u}_j} = 0, \quad (1)$$

where the Lagrangian,  $\mathcal{L} = T - U + W_e$ , is given by the sum of the kinetic energy,  $T$ , the potential energy,  $U$ , and the total piezoelectric energy  $W_e$ . Moreover, the total energy dissipation is described by the total electromechanical dissipation function,  $D$ .

The total potential energy,  $U$ , of the compact multistable system can be viewed by the equivalent manifestation of the elastic energy from the structure and the magnetic energy from the magnetic interactions. These interactions can be qualitatively described by a polynomial function as described below:

$$U = \sum_{i=1}^n \left\{ \sum_{j=0}^{N-1} \left[ -\frac{(-1)^{j+N}}{2(j+1)} a_{j+1}^{(i)} z_i^{2(j+1)} \right] + \frac{1}{2} k_i (z_i - z_{i-1})^2 \right\}, \quad z_0 = 0, \quad (2)$$

where  $n = 2$  is the number of mechanical degrees-of-freedom of the system,  $N^2$  is a number that represents the maximum achievable number of stable positions,  $k_i$  is the equivalent stiffness of each beam, and  $a_j^{(i)}$  are constant polynomial coefficients of the potential energy function. These coefficients are associated with each DoF,  $i$ , and are determined by a polynomial fitting of experimental data or more sophisticated magnetic force models (Rezaei *et al.*, 2021).

The remaining energy terms are detailed in (Costa and Savi, 2024; Costa *et al.*, 2024), resulting in the electromechanical equations of the system given by Eqs. (3) to (6), where the term  $(t)$  that indicated time dependency is conveniently suppressed.

$$m_1 \ddot{z}_1 + c_1 \dot{z}_1 - c_2 (\dot{z}_2 - \dot{z}_1) - \frac{\partial U}{\partial z_1} - \theta_1 v_1 + \theta_2 v_2 = -m_1 \ddot{z}_b, \quad (3)$$

$$m_2 \ddot{z}_2 + c_2 (\dot{z}_2 - \dot{z}_1) - \frac{\partial U}{\partial z_2} - \theta_2 v_2 = -m_2 \ddot{z}_b, \quad (4)$$

$$C_{p1} \dot{v}_1 + \frac{v_1}{R_1} + \theta_1 \dot{z}_1 = 0, \quad (5)$$

$$C_{p2} \dot{v}_2 + \frac{v_2}{R_2} + \theta_2 (\dot{z}_2 - \dot{z}_1) = 0, \quad (6)$$

A non-dimensionalization procedure is performed in Eqs. (3) to (6), using  $\omega_1 = \sqrt{k_1/m_1}$  as reference frequency,  $\tau = \omega_1 t$  as dimensionless time,  $L$  as reference length, and  $V$  as reference voltage. The procedure results in the dimensionless

electromechanical equations defined by Eqs. (7) to (10), where the new resulting dimensionless parameters are presented in Table 2.2

$$\ddot{z}_1 + 2\zeta_1 \dot{z}_1 - 2\zeta_2 (\dot{z}_2 - \dot{z}_1) - \frac{\partial \bar{U}}{\partial \bar{z}_2} - \chi_1 \bar{v}_1 + \chi_2 \bar{v}_2 = -\ddot{z}_b, \quad (7)$$

$$\rho \ddot{z}_2 + 2\zeta_2 (\dot{z}_2 - \dot{z}_1) - \frac{\partial \bar{U}}{\partial \bar{z}_2} - \chi_2 \bar{v}_2 = -\rho \ddot{z}_b, \quad (8)$$

$$\dot{v}_1 + \varphi_1 \bar{v}_1 + \kappa_1 \dot{z}_1 = 0, \quad (9)$$

$$\dot{v}_2 + \varphi_2 \bar{v}_2 + \kappa_2 \dot{z}_2 = 0. \quad (10)$$

Furthermore, the corresponding dimensionless version of the potential energy of the system can be written as:

$$\bar{U} = \sum_{i=1}^n \left\{ \sum_{j=0}^{N-1} \left[ -\frac{(-1)^{j+N}}{2(j+1)} \alpha_{j+1}^{(i)} \bar{z}_i^{2(j+1)} \right] + \frac{1}{2} \prod_{k=2}^i \left[ \rho_{k,k-1} \Omega_{k,k-1}^2 (\bar{z}_i - \bar{z}_{i-1})^2 \right] \right\}, \quad \bar{z}_0 = 0, \quad (11)$$

where  $\rho_{k,k-1} = m_k/m_{k-1}$  and  $\Omega_{k,k-1} = \Omega_k/\Omega_{k-1}$  denote the equivalent mass ratio and the natural frequency ratio, respectively, between two consecutive degrees-of-freedom,  $k-1$  and  $k$ .

## 2.2 Performance Metrics

Harvester performance is evaluated through electrical output variables, considering both instantaneous and average quantities. For a resistive circuit, the instantaneous electrical power is represented by Eq. (12).

$$P_i = \frac{v_i^2}{R_i}, \quad i = 1, 2. \quad (12)$$

The average electrical power is represented by Eq. (13),

$$P_{avg} = \sum_{i=1}^2 \left[ \frac{1}{t_f - t_0} \int_{t_0}^{t_f} P_i dt \right] = \sum_{i=1}^2 \left[ \frac{(v_i^{RMS})^2}{R_i} \right], \quad (13)$$

where  $\square^{RMS}$  denotes the root-mean-square value of a quantity over the interval from a initial time,  $t_0$ , to a final time,  $t_f$ .

Based on Table 2.2 the dimensionless average electrical output power is determined by Eq. (14).

$$\bar{P}_{avg} = \sum_{i=1}^2 \left[ \frac{1}{\tau_f - \tau_0} \int_{\tau_0}^{\tau_f} \bar{P}_i d\tau \right] = \sum_{i=1}^2 \left[ \varphi_i (\bar{v}_i^{RMS})^2 \right]. \quad (14)$$

Here  $\tau_0$  and  $\tau_f$  represents the initial and final dimensionless times, respectively.

## 3. NUMERICAL SIMULATIONS

In this section, numerical simulations are carried out. First, a stability analysis is performed to show different characteristics of each harvester analyzed. Then, a dynamical analysis is performed to compare the performance of each system.

### 3.1 Stability Analysis

The multistability of the system is a interesting point to be investigated since it is directly related to the enhancement of the energy harvesting capacity. This section dedicates a special attention to this issue. The stability analysis is performed combining three steps: the identification of equilibria, the determination of the nature of each equilibrium point, and the evaluation of the potential energy function in a parameter space that comprises all equilibria.

The equilibrium configurations can be determined by identifying the solutions for  $\bar{\mathbf{q}} = \mathbf{f}(\bar{\mathbf{q}}) = \mathbf{0}$ , where  $\bar{\mathbf{q}} = [\bar{z}_1(\tau), \bar{z}_2(\tau), \bar{z}_2(t), \bar{v}_1(\tau), \bar{v}_2(\tau)]$  is the state space vector, and  $\mathbf{f}(\bar{\mathbf{q}})$  are the electromechanical equations in canonical form.

The nature of each equilibrium position are determined through a linearization around each point, evaluating the Jacobian matrix  $\mathbf{J}$ , displayed in Eq. (15).

$$\mathbf{J} = \nabla^T \mathbf{f}(\bar{\mathbf{q}}) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 - \mathcal{F}_1 - \rho_{2,1} \Omega_{2,1}^2 & -2(\zeta_1 + \zeta_2) & \rho_{2,1} \Omega_{2,1}^2 & 2\zeta_2 & \chi_1 & -\chi_2 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \Omega_{2,1}^2 & \frac{2\zeta_2}{\rho_{2,1}} & -\frac{\mathcal{F}_2 + \rho_{2,1} \Omega_{2,1}^2}{\rho_{2,1}} & -\frac{2\zeta_2}{\rho_{2,1}} & 0 & \frac{\chi_2}{\rho_{2,1}} \\ 0 & -\kappa_1 & 0 & 0 & -\varphi_1 & 0 \\ 0 & \kappa_2 & 0 & -\kappa_2 & 0 & -\varphi_2 \end{bmatrix} \quad (15)$$

Tabela 1: **System parameters, definitions and values used in the analysis.**

| Parameter description                                                   | Symbol            | Definition                     | Value                 |
|-------------------------------------------------------------------------|-------------------|--------------------------------|-----------------------|
| Linearized natural frequency of 1st beam                                | $\omega_1$        | $\sqrt{k_1/m_1}$               | -                     |
| Linearized natural frequency of 2nd beam                                | $\omega_2$        | $\sqrt{k_2/m_2}$               | -                     |
| Dimensionless time                                                      | $\tau$            | $\omega_1 t$                   | -                     |
| Dimensionless displacement of the 1st beam                              | $\bar{z}_1(\tau)$ | $z_1(t)/L$                     | -                     |
| Dimensionless displacement of the 2nd beam                              | $\bar{z}_2(\tau)$ | $z_2(t)/L$                     | -                     |
| Dimensionless voltage in the 1st circuit                                | $\bar{v}_1(\tau)$ | $v_1(t)/V$                     | -                     |
| Dimensionless voltage in the 2nd circuit                                | $\bar{v}_2(\tau)$ | $v_2(t)/V$                     | -                     |
| Dimensionless base excitation frequency                                 | $\Omega$          | $\omega/\omega_1$              | $0.01 \rightarrow 10$ |
| Dimensionless base excitation amplitude                                 | $\gamma$          | $A/L$                          | $0.01 \rightarrow 1$  |
| Dimensionless base excitation displacement                              | $\bar{z}_b(\tau)$ | $\gamma \sin(\Omega\tau)$      | -                     |
| Ratio between equivalent masses of 1st and 2nd beams                    | $\rho_{2,1}$      | $m_2/m_1$                      | 1                     |
| Dimensionless mechanical dissipation coefficient of the 1st beam        | $\zeta_1$         | $c_1/(2m_1\omega_1)$           | 0.025                 |
| Dimensionless mechanical dissipation coefficient of the 2nd beam        | $\zeta_2$         | $c_2/(2m_1\omega_1)$           | 0.025                 |
| Ratio between linearized natural frequencies of outer and inner beams   | $\Omega_{2,1}$    | $\omega_2/\omega_1$            | 0.25                  |
| 1st Dimensionless restitution coefficient of the 1st beam               | $\alpha_1^{(1)}$  | $a_1^{(1)}/(m_1\omega_1^2)$    | $-0.5 \rightarrow 2$  |
| 1st dimensionless restitution coefficient of the 2nd beam               | $\alpha_1^{(2)}$  | $a_1^{(2)}/(m_1\omega_1^2)$    | $0.5 \rightarrow 1$   |
| 2nd dimensionless restitution coefficient of the 1st beam               | $\alpha_2^{(1)}$  | $a_2^{(1)}L^2/(m_1\omega_1^2)$ | $1 \rightarrow 5.56$  |
| 2nd dimensionless restitution coefficient of the 2nd beam               | $\alpha_2^{(2)}$  | $a_2^{(2)}L^2/(m_1\omega_1^2)$ | $1 \rightarrow 5.56$  |
| 3rd dimensionless restitution coefficient of the 1st beam               | $\alpha_3^{(1)}$  | $a_3^{(1)}L^4/(m_1\omega_1^2)$ | $1 \rightarrow 6.83$  |
| 3rd dimensionless restitution coefficient of the 2nd beam               | $\alpha_3^{(2)}$  | $a_3^{(2)}L^4/(m_1\omega_1^2)$ | $1 \rightarrow 6.83$  |
| 4th dimensionless restitution coefficient of the 1st beam               | $\alpha_4^{(1)}$  | $a_4^{(1)}L^6/(m_1\omega_1^2)$ | 2.26                  |
| 4th dimensionless restitution coefficient of the 2nd beam               | $\alpha_4^{(2)}$  | $a_4^{(2)}L^6/(m_1\omega_1^2)$ | 2.26                  |
| 1st dimensionless coupling coefficient in the 1st piezoelectric element | $\chi_1$          | $\theta_1 V/(k_1 L)$           | 0.05                  |
| 1st dimensionless coupling coefficient in the 2nd piezoelectric element | $\chi_2$          | $\theta_2 V/(k_1 L)$           | 0.05                  |
| 2nd dimensionless coupling coefficient in the 1st piezoelectric element | $\kappa_1$        | $\theta_1 L/(C_{p1} V)$        | 0.5                   |
| 2nd dimensionless coupling coefficient in the 2st piezoelectric element | $\kappa_2$        | $\theta_2 L/(C_{p2} V)$        | 0.5                   |
| Dimensionless electrical conductance in the 1st circuit                 | $\varphi_1$       | $1/(C_{p1} R_1 \omega_1)$      | 0.05                  |
| Dimensionless electrical conductance in the 2st circuit                 | $\varphi_1$       | $1/(C_{p2} R_2 \omega_1)$      | 0.05                  |
| Dimensionless instantaneous electrical output power in the 1st circuit  | $\bar{P}_1(\tau)$ | $P_1(t)/(C_{p1} \omega_1 V^2)$ | -                     |
| Dimensionless instantaneous electrical output power in the 2nd circuit  | $\bar{P}_2(\tau)$ | $P_2(t)/(C_{p2} \omega_1 V^2)$ | -                     |

$$\mathbf{J} = \nabla^T \mathbf{f}(\bar{\mathbf{q}}) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\partial^2 \bar{U}}{\partial \bar{z}_1^2} & -2(\zeta_1 + \zeta_2) & \rho_{2,1}\Omega_{2,1}^2 & 2\zeta_2 & \chi_1 & -\chi_2 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ \Omega_{2,1}^2 & \frac{2\zeta_2}{\rho_{2,1}} & -\frac{1}{\rho_{2,1}} \left( \frac{\partial^2 \bar{U}}{\partial \bar{z}_2^2} \right) & -\frac{2\zeta_2}{\rho_{2,1}} & 0 & \frac{\chi_2}{\rho_{2,1}} \\ 0 & -\kappa_1 & 0 & 0 & -\varphi_1 & 0 \\ 0 & \kappa_2 & 0 & -\kappa_2 & 0 & -\varphi_2 \end{bmatrix} \quad (16)$$

where  $\mathcal{F}_i = \sum_{j=0}^{N-1} \left[ -(2j+1)(-1)^{j+N} \alpha_{j+1}^{(i)} \bar{z}_i^{2j} \right], (i = 1, 2)$ .

The stability characteristics of each point is evaluated from the eigenvalues of  $\mathbf{J}$ ,  $\mu_j (j = 1, \dots, 6)$ . These points can be classified into three sets:

1. Stable if  $\{\mu_j \in \mathbb{C} \mid \text{Re}(\mu_j) < 0, \forall j\}$ ;
2. Unstable if  $\{\mu_j \in \mathbb{C} \mid \text{Re}(\mu_j) > 0, \exists j\}$ ;
3. Center if  $\{\mu_j \in \mathbb{C} \mid \text{Re}(\mu_j) = 0, \exists j\}$ .

Additionally, according to the Hartman-Grobman theorem, the stability of the linearized system at the vicinity of an equilibrium point corresponds to the nonlinear system as long the point is hyperbolic, meaning that there is no eigenvalue that the real part is zero ( $\text{Re}(\mu_j) \neq 0, \forall j$ ) (Junior *et al.*, 2016). On this basis, for this case all found solutions are hyperbolic, and the unstable points can be split into two distinct groups: saddle-type unstable points and source-type unstable points. Saddle-type unstable points exhibit one positive eigenvalue, indicating the presence of a single unstable direction in the state space. Alternatively, source-type unstable points exhibit two positive eigenvalues, indicating the presence of two unstable directions.

Tabela 2: **System parameters, definitions and values used in the analysis.**

| Configuration | N | Polynomial Parameters |                  |                  |                  |                  |                  |                  |                  |
|---------------|---|-----------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|               |   | $\alpha_1^{(1)}$      | $\alpha_2^{(1)}$ | $\alpha_3^{(1)}$ | $\alpha_4^{(1)}$ | $\alpha_1^{(2)}$ | $\alpha_2^{(2)}$ | $\alpha_3^{(2)}$ | $\alpha_4^{(2)}$ |
| 4–Stable      | 2 | 2                     | 1                | -                | -                | 1                | 1                | -                | -                |
| 9–Stable      | 3 | -0.5                  | 1.7              | 1                | -                | 0.5              | 1.7              | 1                | -                |
| 16–Stable     | 4 | 2                     | 5.56             | 6.83             | 2.26             | 1                | 5.56             | 6.83             | 2.26             |

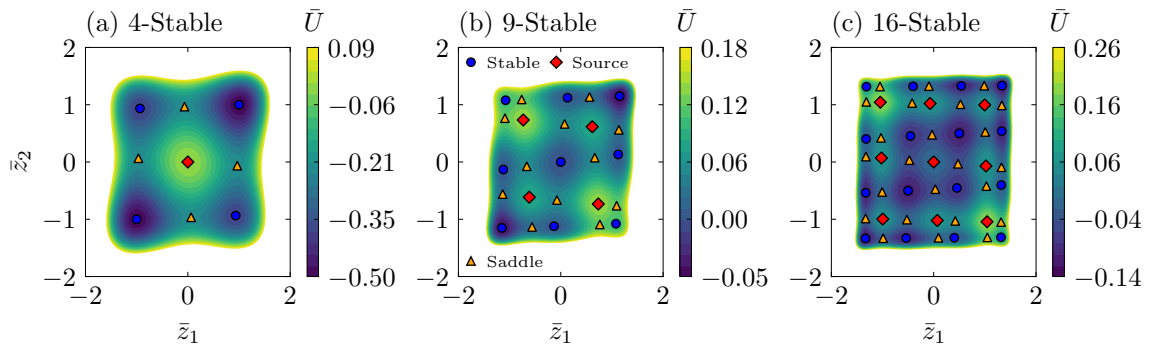


Figura 3: **Equilibrium configurations considering  $\Omega_{2,1} = 0.25, \rho_{2,1} = 1$ . (a) A 4-Stable configuration, (b) A 9-Stable configuration, and (c) a 16-Stable configuration.**

The solution of the problem  $\dot{\mathbf{q}} = \mathbf{f}(\bar{\mathbf{q}}) = \mathbf{0}$  yields equilibrium positions of the form  $\{\bar{z}_1, \dot{\bar{z}}_1, \bar{z}_2, \dot{\bar{z}}_2, \bar{v}_1, \bar{v}_2\} = \{\bar{Z}_1, 0, \bar{Z}_2, 0, 0, 0\}$ , which can be visualized in the reduced plane subset  $\bar{z}_1, \bar{z}_2 = \bar{Z}_1, \bar{Z}_2$ . Figure 3 presents the results of the stability analysis by considering different sets of polynomial coefficients, as listed in Table 2, with structural parameters of  $\Omega_{2,1} = 0.25$  and  $\rho = 1$ . Corresponding to the magnetic arrangements in Figure 1(a) to 1(c), which generates the 4-Stable, 9-Stable and the 16-Stable configurations, respectively. The color bars indicate potential energy levels, with darker tones representing lower energies and lighter tones representing higher energies. Blue circles represent Stable equilibria, while orange triangles and red diamonds represent unstable saddle, and unstable source equilibria, respectively.

The 4-Stable configuration displays 9 equilibria, which allows smoother transitions between a lower energy level to a higher energy level, but maintains a high potential energy barrier. Alternatively, the 9-Stable and 16-Stable configurations show more abrupt changes in energy levels due to the increased number of equilibrium positions. These abrupt changes occur at values around  $\{\bar{z}_1, \bar{z}_2\} = \{-1.6, 1.3\}$  and  $\{\bar{z}_1, \bar{z}_2\} = \{1.6, -1.3\}$ . Alternatively, in the remaining regions, it is possible to observe a reduction in potential energy barrier, which can be beneficial for energy harvesting purposes.

### 3.2 Dynamics and Performance Evaluation

In this section, a performance analysis is conducted for each established configuration. The analysis is presented in the form of two-dimensional diagrams that maps the dynamics and performance of the system within a specific range of parameter values. Specifically, the excitation parameter domain,  $\gamma \times \Omega$ , is chosen as it represents the variations in the harvester's operational conditions.

The diagrams are built with a grid of  $200 \times 200$  points, each of which is obtained from an explicit fourth-order Runge-Kutta scheme integration considering a  $N_T = 4000$  excitation periods, and a time-step of  $\Delta\tau = 2\pi/(\delta\Omega)$ . Here,  $\delta$  is the number of time-steps each period of mechanical excitation contains, so that the total number of time steps is  $N_{\Delta\tau} = \delta N_T$ , and the final time of integration in each point is determined by  $\tau_f = 2/pi N_T/\Omega$ .

Figure 4 summarizes the results for the three configurations (one per column). The top row shows Dynamical Response Diagrams (DRDs), which classifies the dynamical response type the system manifests at each point. Periodic responses are labelled as  $kT$ , indicating that the system repeats its trajectory every  $k$  excitation periods (T). Alternatively, MP classification (light blue) denotes dynamical responses with period greater than  $5T$  or quasi-periodic behavior. CH and HC represent chaotic and hyperchaotic responses, respectively.

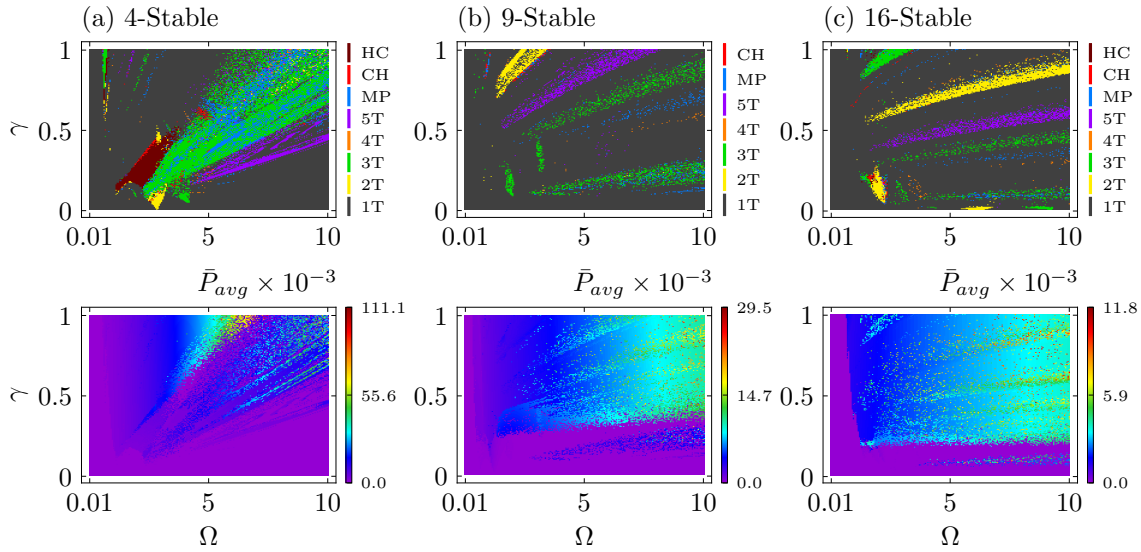


Figura 4: **Dynamical Response Diagrams (DRDs) at the top and Output Power Diagrams (OPDs) at the bottom for (a) the 4-Stable configuration, (b) the 9-Stable configuration, and (c) the 16-stable configuration. Colorbars indicate dynamical responses (at top), and  $\bar{P}_{avg} \times 10^{-3}$  (at bottom).**

DRDs are important as they map the dynamics of the system within the chosen parameter space. In this case, the dynamics can be associated with performance metrics. In this case, they indicate that 9-Stable and 16-Stable configurations manifests predominantly periodic dynamics, while the 4-Stable configuration exhibit a large clear hyperchaotic region and smaller chaotic islands. In all cases, 4T responses are the least frequent, while 1T, 2T, 3T and 5T responses are more common. MP dynamics is also more frequent in the 4-Stable configuration, while is rare in the remaining ones.

The bottom row of Figure 4 presents Output Power Diagrams (OPDs), which map the average output power,  $\bar{P}_{avg}$  over the same parameter space. Comparison of the OPDs shows that increasing the number of stable equilibria expands the operational bandwidth and improves the consistency of high output power, particularly for higher frequencies. In the 4-stable configuration, higher output power is manifested mainly at low-to-medium frequencies, while at higher frequencies, high-power points are interspersed with low-power points, producing an intermittent pattern. In the 9-stable configuration, the high-performance region becomes broader and more coherent is larger and clearer, maintaining high output power in low-to-medium frequencies, and indicating more consistency in power conversion in higher frequencies. The 16-stable configuration follows the same trend, with a slightly larger high-performance area.

On the other hand, the maximum output power decreases drastically as the number of stable positions increases for the tested configurations. The peak power from the 4-Stable to the 9-Stable configuration decreases by approximately 73.4%, and from the 9-Stable to the 16-Stable configuration by about 60%. These results indicate the increasing the number of stable positions does not necessarily improve overall performance. In general, performance improves at lower excitation amplitudes and higher frequencies, whereas gains at higher excitation amplitudes are not guaranteed. Further investigation is required to determine whether this behavior is intrinsic to increasing number of equilibria or specific to the selected configurations.

#### 4. CONCLUSIONS

This work addresses the generalization of multistability in a compact multistable energy harvester (CMEH). The proposed design consists of a cut-out outer beam and an inner beam, each with a tip mass and an attached piezoelectric element, coupled through magnetic interactions generated by magnets arranged around the tip masses. Different stability conditions are obtained by changing the magnetic setup.

Based on this concept, an electromechanical lumped model is developed using an energy-based formulation. A general polynomial expression for the potential energy is introduced, combining linear stiffness and polynomial terms to represent the equivalent interaction between magnetic and elastic forces. A dimensionless electromechanical model is then obtained to analyze the qualitative behavior of the system.

A stability analysis of the dimensionless model shows that the proposed energy formulation captures different multistability conditions. Three representative configurations are selected, with 4, 9, and 16 stable equilibrium positions.

Dynamic and performance analyses of these configurations show that increasing the number of stable positions can

broaden the operational bandwidth. At low excitation amplitudes, this shift favors higher-power operating regimes, and at high amplitudes it tends to improve power-conversion consistency. However, for the tested configurations, increasing the number of stable positions also leads to a substantial reduction in maximum output power, which is a relevant drawback for energy conversion. Further investigation is needed to determine whether these trends are configuration-specific or represent a general pattern of the system.

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